

Rui Fang

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Ph.D. Candidate, Graduate Institute of Communication Engineering, National Taiwan University

rfang@arbor.ee.ntu.edu.tw +886 905 001 986 [rui-fang.github.io](https://github.com/rui-fang)

[Google Scholar](#) [GitHub](#) [ORCID](#) [LinkedIn](#)

Education

National Taiwan University 2021–Present

Ph.D. Candidate, Graduate Institute of Communication Engineering

Network Database Laboratory. Advisor: Prof. Ming-Syan Chen.

National Taipei University of Technology 2018–2021

M.S., Graduate Institute of Manufacturing Technology

Advanced Microsystems and Device Laboratory. Advisor: Prof. Chih-Cheng Lu.

National Taipei University of Technology 2014–2018

B.S., Mechanical Engineering

College of Mechanical and Electrical Engineering.

Research Interests

Efficient inference and model compression for Transformers: early exiting, quantization, structured pruning, KV-cache management, and looped language models. Related work includes multi-task learning and unlearning, few-shot adaptation under distribution shifts, secure on-device inference, and interpretable biomedical AI.

Technical Skills

Programming	Python, C++, CUDA, $\text{\LaTeX}$.
Frameworks	PyTorch, TensorFlow, JAX.
Methods	Quantization, pruning, early exiting, LoRA/parameter-efficient fine-tuning, knowledge distillation, and KV-cache optimization.
Systems	GPU programming, on-device inference, and trusted execution environments.

Awards and Academic Service

Award: National Taiwan University Wen-Tzu Hsiang Memorial Scholarship, 2025.

Conference reviewing: CVPR, ICML, IJCAI, ECAI, ECCV, ICLR, and NeurIPS.

Publications

*Equal contribution (co-first authors). Accepted papers are explicitly marked; manuscripts under review and preprints are listed separately.

Peer-reviewed and Accepted Papers

- [1] **Amortized-Precision Quantization for Early-Exit Vision Transformers**
Rui Fang, Hsi-Wen Chen, and Ming-Syan Chen
NeurIPS 2026. Accepted. [\[Paper\]](#)
- [2] **KV Admission: Learning What to Write for Efficient Long-Context Inference**
Yen-Chieh Huang, Pi-Cheng Hsiu, Rui Fang, and Ming-Syan Chen
EMNLP 2026. Accepted. [\[Paper\]](#) [\[Code\]](#)

- [3] **Mask as Memory: Streaming Multi-task Pruning for Vision Transformers**
Ying-Hua Huang, **Rui Fang**, Hsi-Wen Chen, and Ming-Syan Chen
ECML PKDD 2026, Research Track, pp. 391–409. [\[Paper\]](#)
- [4] **HSMLog: Small Language Model-Assisted Hardware Security Module Log Anomaly Detection with Behavioral Analysis**
Chia-Hsuan Wu, Dar-Hsin Dustin Wu, **Rui Fang**, Yi-Ting Lee, Chia-Chih Lin, and Ming-Syan Chen
ISSRE 2026, Industry Track. Accepted. [\[Paper\]](#)
- [5] **LoGIC: Multi-LoRA Guided Importance Consensus for Multi-Task Pruning in Vision Transformers**
Yu-Hong Chou*, **Rui Fang***, Hsi-Wen Chen, and Ming-Syan Chen
AAAI 2026, 40(25):20588–20596. [\[Paper\]](#)
- [6] **Dual Alignment Framework for Few-shot Learning with Inter-Set and Intra-Set Shifts**
Siyang Jiang, **Rui Fang**, Hsi-Wen Chen, Wei Ding, Guoliang Xing, and Ming-Syan Chen
NeurIPS 2025; Advances in Neural Information Processing Systems 38, pp. 64804–64830 (*proceedings record: 2026*). [\[Paper\]](#) [\[Code\]](#)
- [7] **SEAL: Secure and Efficient Adaptive Layering for On-Device Language Models with TEE**
Wen-Tzu Chang, **Rui Fang**, and Ming-Syan Chen
IEEE EDGE 2025, pp. 138–143. [\[Paper\]](#)
- [8] **BiLEE: Bi-Level Early Exiting for Generative Document Retrieval**
Rui Fang, Chin-Yuan Yeh, Hsi-Wen Chen, and Ming-Syan Chen
ECAI 2024, pp. 4035–4042. [\[Paper\]](#)
- [9] **Dual-Triangular QR Decomposition with Global Acceleration and Partially Q-Rotation Skipping**
Rui Fang, Siyang Jiang, Hsi-Wen Chen, Wei Ding, and Ming-Syan Chen
ICFPT 2022, pp. 1–4. [\[Paper\]](#)
- [10] **A visually interpretable detection method combines 3-D ECG with a multi-VGG neural network for myocardial infarction identification**
Rui Fang, Chih-Cheng Lu, Cheng-Ta Chuang, and Wen-Han Chang
Computer Methods and Programs in Biomedicine, 219:106762, 2022. [\[Paper\]](#)

Manuscript Under Review

- [11] **LOOPQ: QUANTIZATION FOR LOOPED LANGUAGE MODELS**
Rui Fang, Hsi-Wen Chen, and Ming-Syan Chen
Submitted to ICLR 2027; under review. Manuscript, 2026.

Preprint

- [12] **Interference-Aware Multi-Task Unlearning**
Ying-Hua Huang, **Rui Fang**, Hsi-Wen Chen, and Ming-Syan Chen
arXiv:2605.19042, 2026. [\[Paper\]](#)